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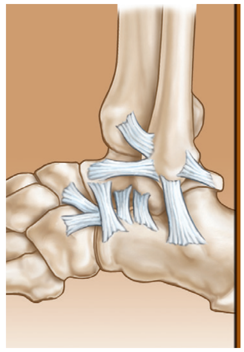

Imaging of the Foot and Ankle

Anish R. Kadakia, Amiethab A. Aiyer

RADIOGRAPHS

Radiographs are often the initial imaging study performed for any patient presenting with a problem related to the foot and ankle. Radiographs are clearly beneficial when evaluating for suspected arthritis, osteonecrosis, tumors, nonunion, or trauma. In addition, the alignment of the ankle and foot may contribute to the presenting complaint even if the radiographs being viewed are free of pathologic abnormalities. To appropriately evaluate the osseous structures and alignment of the foot, the radiographs should be obtained with weight bearing in all cases, except when a fracture may be suspected (Figs. 112.1–112.3). Specifically, in the setting of a suspected Lisfranc injury, if the initial non–weight-bearing radiographs are negative, follow-up weight-bearing films should be ordered to evaluate for diastasis of the tarsometatarsal joints.

COMPUTED TOMOGRAPHY

The most current generation of multidetector computed tomography (CT) scanners are capable of providing high-resolution images in all imaging planes and are superb for evaluating the extent and location of intra-articular fractures in the area of the ankle and hindfoot. Given the difficulty in assessing joint congruity in the hindfoot, follow-up CT imaging is recommended in this setting to better appreciate the fracture pattern and joint congruity. Cystic osteochondral defects of the talus are excellently visualized with this modality to accurately calculate the size of the lesion and guide preoperative planning. It is also an excellent method of assessing bony healing, including evaluation of fractures and arthrodeses. Sufficient data are available to suggest that determination of the percentage of bony trabeculation across an arthrodesis site in the ankle and hindfoot is grossly underestimated with plain radiography. The administration of intra-articular contrast material followed by CT imaging through the ankle (CT arthrography) is an accurate means of assessing internal derangement in patients who have a contraindication to magnetic resonance imaging (MRI). CT technology has now evolved into obtaining weight bearing scans of the lower extremity which can provide the surgeon with functional information regarding joint biomechanics. This technology can be extremely useful in situations where radiographs are not able to provide information due to overlap of bones. For example, in patients with midfoot arthrosis, weight-bearing CT is able to isolate each joint in the midfoot and assess for bone-on-bone apposition during weight bearing. It also provides accuracy and ease of assessing syndesmotic alignment as well as degree of subfibular impingement in hindfoot valgus deformity.

ULTRASOUND

The primary advantage of ultrasound compared with MRI is the ability to evaluate the tendinous structures in a dynamic fashion.[1] Ultrasound is the imaging modality of choice for the evaluation of peroneal tendon subluxation. The evaluation for tendon apposition in the setting of an Achilles tendon rupture or for evaluation of flexor hallucis longus (FHL) stenosing tenosynovitis are examples of when the ability to perform a dynamic examination is critical. Advantages include a relative cost saving compared with MRI and improved localization of soft tissue structures for injection. A limitation of ultrasound is that visualization of the affected structure may prove more difficult in patients with a high amount of subcutaneous fat.[2] Additionally, the clinician is reliant upon the interpretation of the radiologist and the skill of the ultrasonographer for the determination of the pathologic findings. Good communication between the clinician and the radiologist is critical to ensure that the radiologist has an understanding of the exact clinical concern so the examination can be tailored appropriately.

MAGNETIC RESONANCE IMAGING

MRI is well accepted as the primary noninvasive imaging modality for providing a global assessment of the foot and ankle and is very accurate for the evaluation of cartilaginous, tendon, ligament, and bony abnormalities. MR arthrography of the ankle can be used to obtain a more precise evaluation of articular cartilage injuries to the talar dome or tibial plafond.[3] T1-weighted fat-saturated images are key to evaluate the gadolinium contrast material in the joint. These images show the contrast to the greatest advantage. Because evaluation of bone marrow is also critical, T1-weighted non–fat-saturated images are also essential. T1-weighted images in general are superior to evaluate the anatomy, bone marrow, hemorrhage, masses, fat, and fatty infiltration of muscles. Proton density or intermediate images are high resolution and therefore important in the evaluation of ligaments, tendons, and small structures such as cartilage.[3] T2-weighted images are the best sequence to evaluate for fluid,

Abstract

As part of the initial evaluation of an athlete with foot or ankle pain, it is important to obtain appropriate imaging. Imaging can include radiographs, magnetic resonance imaging (MRI), computed tomography (CT), or ultrasound. Depending on the differential diagnoses for a patient's source of pain, imaging can be selected that will allow for optimal visualization of structures thought to be implicated. Radiographs and CT scans tend to be more optimal for evaluating osseous structures; CT scans enables clearer evaluation of intra-articular pathology. An MRI is more optimal for evaluating soft tissue structures, including tendons and ligaments. Ultrasound is operator-dependent, but can evaluate tendons in a dynamic fashion.

Keywords

imaging
MRI
CT
ultrasound

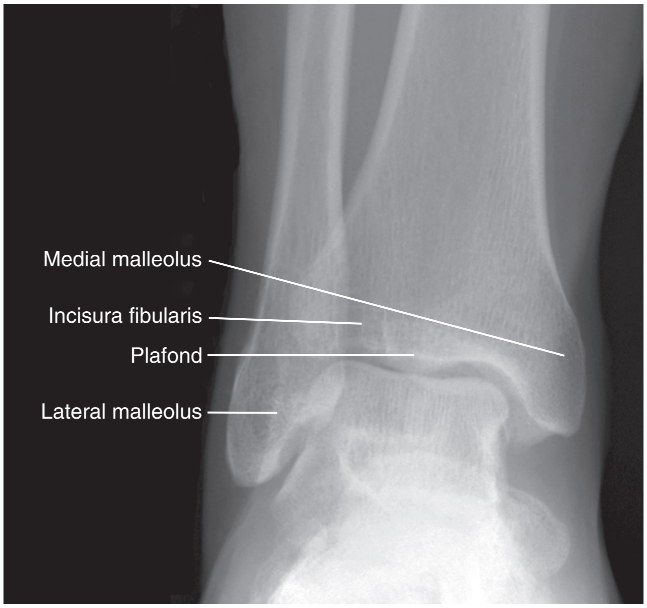

Fig. 112.1 An anteroposterior standing radiograph of the ankle. (From Miller M, Hart J, MacKnight J, eds. *Essential Orthopaedics*. Philadelphia: Elsevier; 2008.)

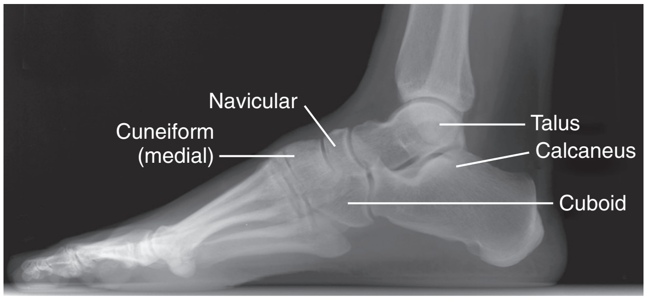

Fig. 112.2 A lateral standing radiograph of the foot. (From Miller M, Hart J, MacKnight J, eds. *Essential Orthopaedics*. Philadelphia: Elsevier; 2008.)

edema, joint effusions, muscle, and soft tissue injuries. Fluid sensitivity is increased with use of fat saturation.[4]

Dedicated foot and ankle coils or smokestack coils are critical in obtaining high-quality imaging of foot and ankle disorders. Images are performed in the sagittal, axial, and coronal imaging planes. When ordering an MRI, the specific area of concern should be communicated to the radiologist. Given that the smallest field of view optimizes image quality, an MRI scan that captures the entire foot and ankle will compromise the quality of the foot images. Additionally, specific protocols may exist for the forefoot versus the hindfoot, and therefore the specific site of the pathologic condition should be stated to ensure that the appropriate MRI protocol is used.

IMAGING OF TENDONS

Given a tendon's lack of fluid, a normal tendon has low signal intensity on all imaging sequences (Fig. 112.4). Axial cuts are the most useful to determine the presence and extent of tenosynovitis and tendinosis. Coronal imaging, although complementary,

Fig. 112.3 An anteroposterior standing radiograph of the foot. (From Miller M, Hart J, MacKnight J, eds. *Essential Orthopaedics*. Philadelphia: Elsevier; 2008.)

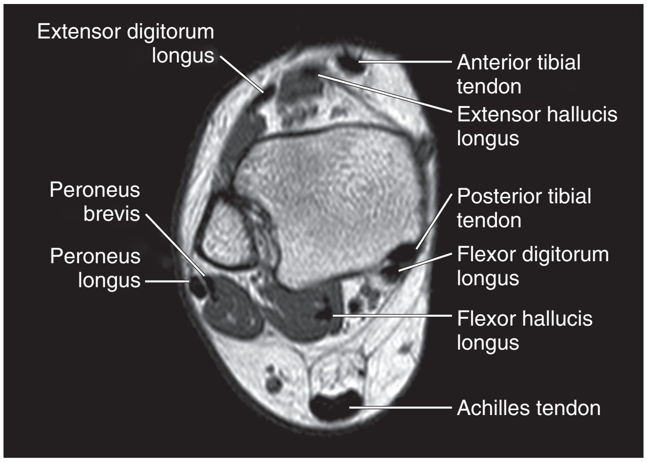

Fig. 112.4 An axial T1-weighted magnetic resonance image of the distal leg. The tendons at this level are denoted.

provides the least information. An area of focal thickening is not a normal finding.

Achilles Tendon

Disorders of the Achilles tendon are diagnosed with relative ease as a result of the subcutaneous nature of the tendon. The presence of tendinosis is noted by focal thickening of the tendon that moves in conjunction with the tendon, as opposed to paratenonitis. The primary utility of MRI is to determine the precise location and affected volume of the tendon for preoperative planning. Increasingly, with nonoperative management of Achilles tendon ruptures demonstrating good results with functional rehabilitation, MRI

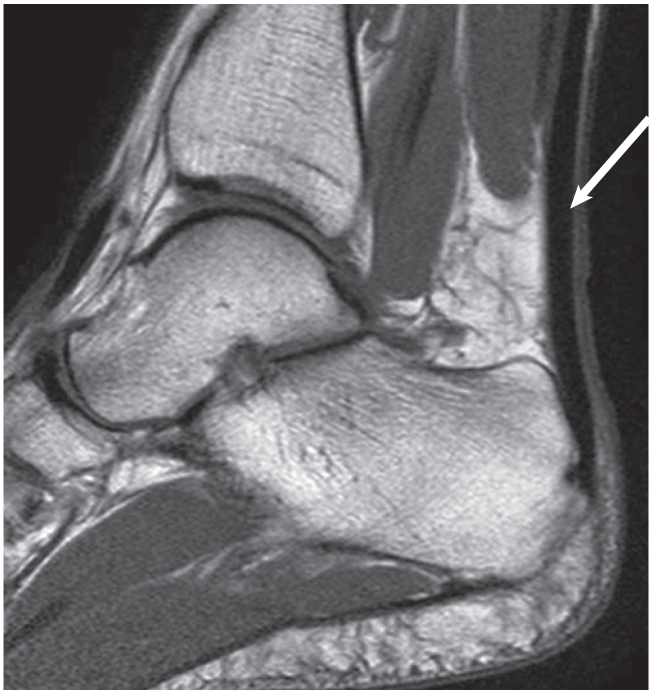

Fig. 112.5 A T1-weighted image of a normal Achilles tendon. Note the uniform thickness and the low signal intensity of the tendon *(arrow)*. (From Joos D, Tran N, Kadakia AR. Achilles tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

is very useful to determine if adequate apposition is obtained with plantar flexion of the ankle after the cast has been placed as an alternative to ultrasonography.

Normal Appearance

The primary evaluation of the Achilles tendon is performed with use of a combination of T1- and T2-weighted axial and sagittal images. The tendon has a near uniform thickness and flattens out distally as it inserts into the calcaneus. The tendon should be taut and parallel when comparing the anterior and posterior margins of the tendon (Fig. 112.5). A flat or concave anterior margin is normal. The normal anteroposterior diameter on axial or sagittal imaging is 7 mm.

Direct Magnetic Resonance Imaging Signs of Disease

Intermediate or high signal intensity on either a T1- or T2-weighted image with continuity of the tendon is indicative of Achilles tendinopathy. Fusiform thickening of the tendon noted on T1-weighted sagittal imaging is diagnostic of Achilles tendinosis (Fig. 112.6). High signal intensity on a T2-weighted image that is as bright as fluid is suggestive of a more acute process and is sometimes referred to as a "partial tear." Calcification of the Achilles insertion is easily visualized on T1-weighted sagittal images and is diagnostic of insertional Achilles tendinopathy. A teardrop fluid signal (high intensity) anterior to the Achilles tendon at the level of the calcaneus is suggestive of retrocalcaneal bursitis (Fig. 112.7). Fluid signal (high intensity) posterior to the Achilles tendon itself is suggestive of retro Achilles bursitis. A fluid signal defect (high intensity) within the Achilles tendon on both T1- and T2-weighted fat-saturated images is indicative of an acute rupture (Fig. 112.8).

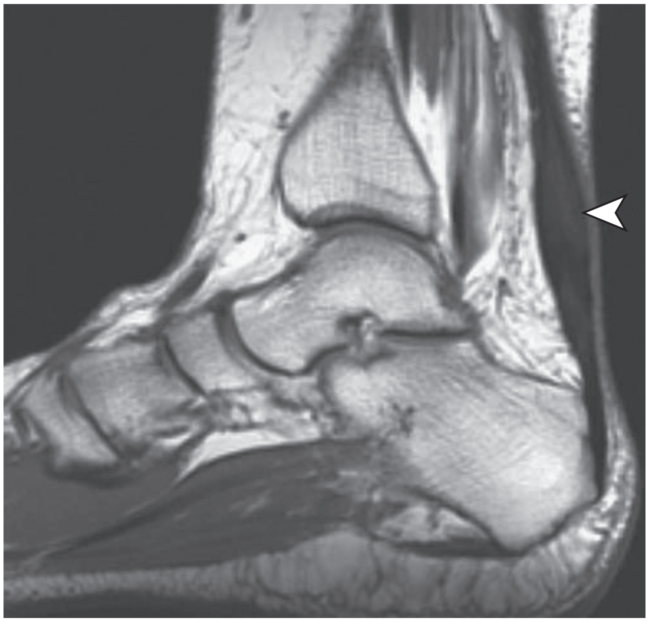

Fig. 112.6 A T1-weighted magnetic resonance image of a patient with noninsertional Achilles tendinosis. Note the intermediate signal intensity and fusiform thickening of the tendon *(arrowhead)*. (From Joos D, Tran N, Kadakia AR. Achilles tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

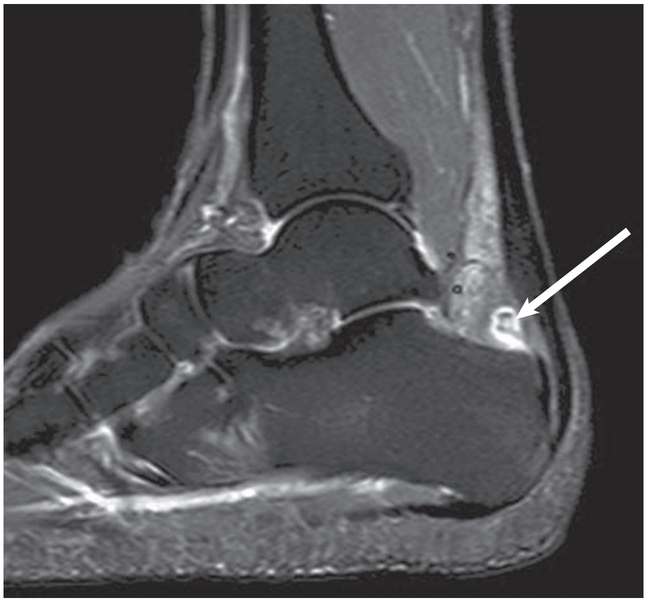

Fig. 112.7 A T2-weighted fat-saturated image with a teardrop high signal intensity immediately anterior to the Achilles at the level of the calcaneus is consistent with retrocalcaneal bursitis *(arrow)*. (From Joos D, Tran N, Kadakia AR. Achilles tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

Indirect Magnetic Resonance Imaging Signs of Disease

High signal intensity on T1- or T2-weighted fat-saturated imaging of the Kager fat pad is suggestive of paratenonitis. A wavy or

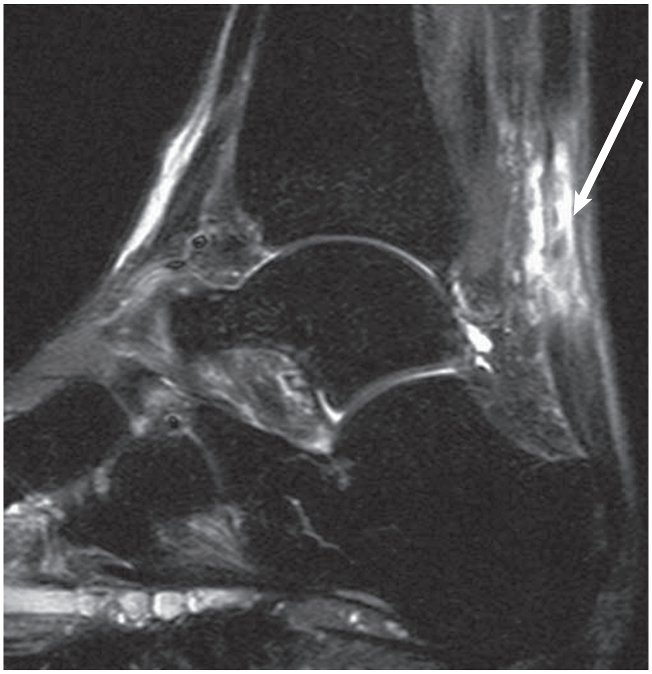

Fig. 112.8 A T2-weighted magnetic resonance image of an acute Achilles rupture. Note the high signal intensity *(arrow)* that completely disrupts the low signal of the Achilles tendon. (From Joos D, Tran N, Kadakia AR. Achilles tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

retracted appearance of the tendon on T1- or T2-weighted fat-saturated imaging is highly suggestive of an Achilles rupture.

Pitfalls

A small amount of increased signal intensity near the insertion of the Achilles tendon represents interposed fat and is normal; it should not be confused with insertional tendinosis. The plantaris tendon can be intact in the presence of an Achilles rupture and should not be confused with a partial tear or intact Achilles tendon.

Posterior Tibial Tendon

Routine use of advanced imaging to determine of the presence of posterior tibial tendon disease is not required. Given the associated flatfoot deformity and subcutaneous position of the tendon, physical examination plays the major role in the diagnosis. However, in cases of suspected tenosynovitis or posttraumatic or sudden-onset flatfoot, and in patients in whom possible debridement and retention of the tendon is being considered, MRI is an excellent adjunct. Isolated debridement and retention of the tendon is typically considered only in the setting of tenosynovitis without intrinsic disease. In the setting of tendinosis, retention of the posterior tibial tendon with an associated tendon transfer will improve the postoperative inversion strength but carries the risk of persistent pain and should be performed with caution.

Magnetic Resonance Imaging

Normal appearance. Axial cuts are the most useful in evaluating the posterior tibial tendon. Ideally the patient should be

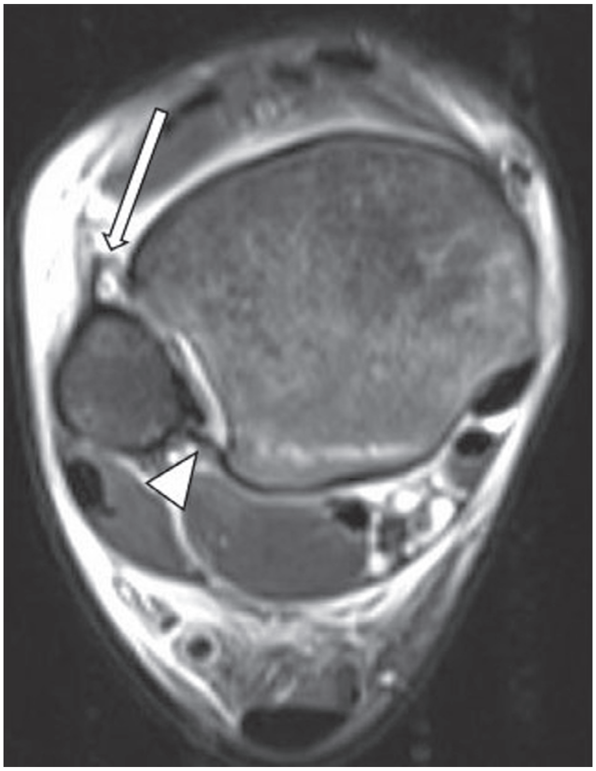

Fig. 112.9 A T2-weighted magnetic resonance image of acute syndesmotic injury. Anteroinferior tibiofibular ligament *(arrow)* is disrupted with surrounding high signal intensity. Posteroinferior tibiofibular ligament *(arrowhead)* is intact with associated nondisplaced posterior distal tibial fracture signifying a high- grade syndesmotic injury. (From Joos D, Sabb B, Tran NK, et al. Acute ankle ligament injuries. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

positioned prone with the foot plantar flexed to place the tendons in a more linear position and reduce the "magic angle" effect. The posterior tibial tendon should be of uniform thickness throughout its length (Fig. 112.9).

Direct magnetic resonance imaging signs. The presence of fluid (high signal intensity) adjacent to the posterior tibial tendon on T2-weighted imaging is sensitive for tenosynovitis (Fig. 112.10). This finding is not specific and is seen in 22% of asymptomatic persons. A focal area of increased signal on T2-weighted images, known as a *punctate signal*, corresponds to a small intrasubstance tear. A focal area of increased signal intensity within the substance of the tendon that extends to the tendon surface on T1-weighted imaging and corresponding high signal intensity on T2-weighted imaging is consistent with a tear.[5] Complete absence of the posterior tibial tendon or replacement of the normal tendon location with high signal intensity on T2-weighted imaging indicates a rupture. Replacement or infiltration of the normally low signal intensity of the tendon with intermediate (gray) signal is consistent with tendinosis (Fig. 112.11). A small tendon caliber (smaller diameter than the flexor digitorum longus) is consistent with atrophic tendinosis and can be seen on any axial imaging sequence.

Indirect magnetic resonance imaging signs. Periretinacular soft tissue edema and thickening of the flexor retinaculum on

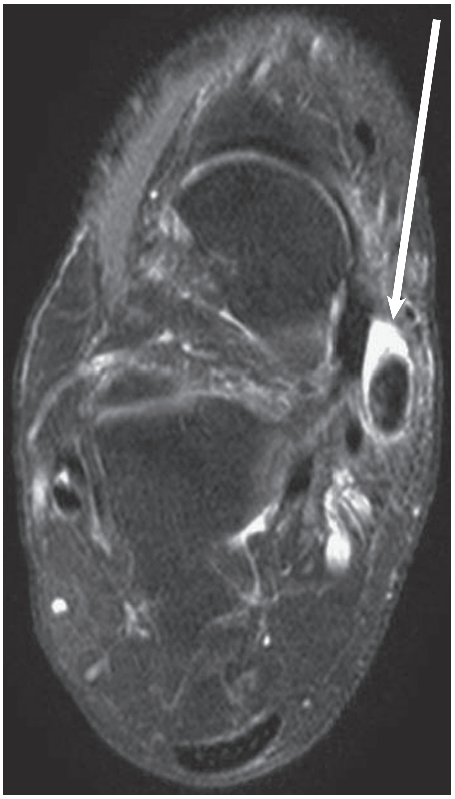

Fig. 112.10 An axial T2-weighted image of a patient with tenosynovitis of the posterior tibial tendon. Note the high signal intensity *(fluid)* adjacent to the tendon *(arrow)*. (From Joos D, Kadakia AR. Flexor tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

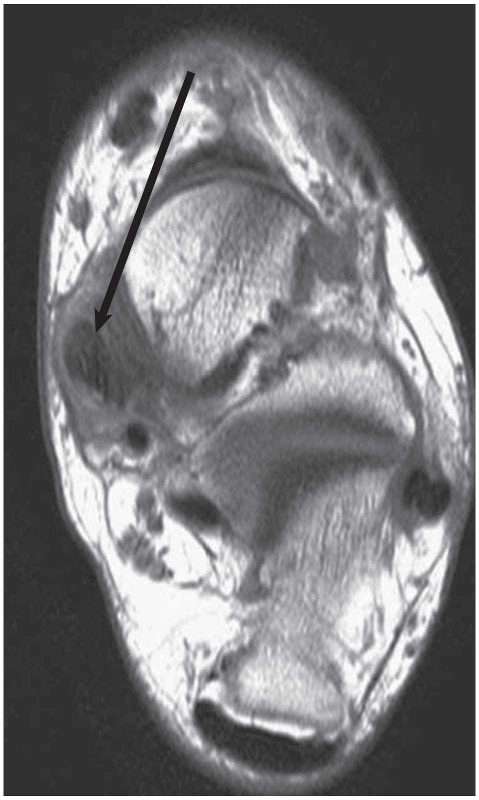

Fig. 112.11 An axial T1-weighted image with infiltration of the normally low signal intensity of the posterior tibial tendon with intermediate *(gray)* signal *(arrow)*. (From Joos D, Kadakia AR. Flexor tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

any imaging sequence is suggestive of disease. Soft tissue edema adjacent to the flexor retinaculum can be seen on T2-weighted imaging. High signal intensity within the medial malleolus without a prior history of trauma may also be seen. High signal intensity noted on T2-weighted imaging within the accessory navicular can be present.

Pitfalls

A small amount of fluid within the tendon sheath can be normal; correlation with intrasubstance increased signal intensity is more specific for tendon disease.

Flexor Hallucis Longus

FHL tendinopathy has classically been described as a disorder in female ballet dancers, in whom constant ankle hyper plantar flexion can cause FHL irritation at its entrance to the flexor retinaculum. Recently it has been noted to affect other patients with chronic foot or ankle pain who often initially were diagnosed with other foot or ankle pathologic conditions. Ultrasound can be effective in determining the presence of dynamic entrapment of the FHL within the fibro-osseous tunnel. MRI is also particularly useful to evaluate for synovitis, the extent of tendon degeneration, and edema within the os trigonum, which can be associated with this condition.

Magnetic Resonance Imaging

Normal appearance. Sagittal cuts are a useful adjunct when evaluating the FHL as it passes under the sustentaculum tali to the forefoot. Ideally the patient should be positioned prone with the foot plantar flexed to place the tendons in a more linear position and reduce the "magic angle" effect. The tendon has a very low-lying muscle belly that is normally present at the level of the ankle.

Direct magnetic resonance imaging signs. MRI findings of tendinosis as outlined previously, with increased signal intensity within the tendon or tears of the tendon, are rarely seen in the FHL. The presence of fluid (high signal intensity) on T2-weighted imaging is diagnostic of FHL tenosynovitis. The most common location is posterior to the talus within the fibro-osseous tunnel (Fig. 112.12). However, it can also occur in the midfoot at the knot of Henry or distally at the level of the first metatarsophalangeal joint as it passes between the sesamoids. Careful inspection of the entire length of the FHL should be performed from the ankle to the insertion into the distal phalanx because compression can occur in the ankle, midfoot, and forefoot. The presence of fluid proximal and distal to the tendon at the level of the fibro-osseous tunnel posteriorly is diagnostic of stenosing tenosynovitis.[6] The severe compression limits the fluid from entering the tunnel.

Indirect magnetic resonance imaging signs. High signal intensity within the os trigonum on T2-weighted imaging should

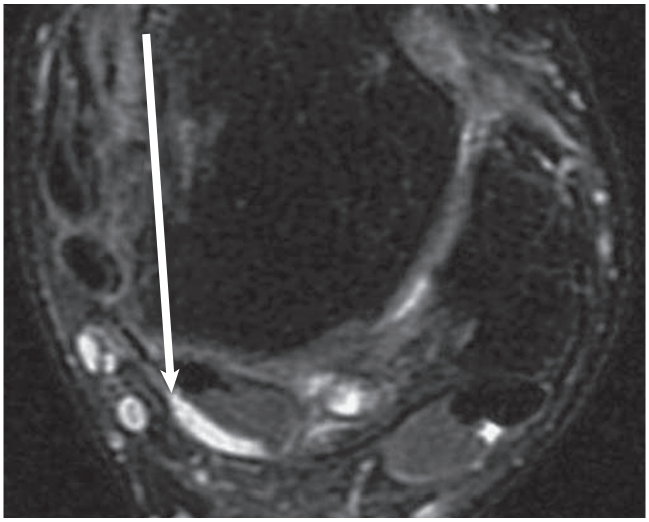

Fig. 112.12 An axial T2-weighted image with high signal intensity *(arrow)* adjacent to the flexor hallucis longus at the posterior aspect of the talus. (From Joos D, Kadakia AR. Flexor tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

prompt close inspection of the FHL because both FHL synovitis and os trigonum syndrome can occur simultaneously.

Pitfalls

Failure to inspect the entire length of the tendon can lead one to miss the less common but symptomatic compression of the FHL within the knot of Henry and sesamoid complex.

Peroneal Tendons

Intrinsic disorders of the peroneal tendons include tenosynovitis, tendinosis, tendon tears, and painful os peroneum syndrome. Additionally, symptomatic subluxation or intrasheath subluxation may be present, producing pain in addition to instability. Chronic subluxation of the tendons increases the risk of developing degenerative tears of the tendons as a result of the mechanical trauma from the posterolateral fibrocartilaginous ridge of the fibula. Ultrasound can be a very effective tool for examination of the peroneal tendons, specifically in the setting of intrasheath subluxation, where subjective "snapping" may be the only appreciable finding. MRI is very useful when considering operative intervention to determine the presence of synovitis, tendinosis, tears, a low-lying muscle belly of the brevis, and a peroneus quartus, in addition to visualizing the concavity of the fibular groove.

Magnetic Resonance Imaging

Normal appearance. The primary evaluation of the peroneal tendons is performed using a combination of T1- and T2-weighted axial images. The peroneus quartus is a normal muscle variant encountered in 13% to 26% of patients. It will appear as a third tendon within the peroneal tendon sheath (Fig. 112.13). The common insertion is the calcaneus, although multiple variants have been noted. A flat or convex fibular groove is associated with peroneal tendon disorders, but a flat groove is seen in more than two thirds of normal persons.

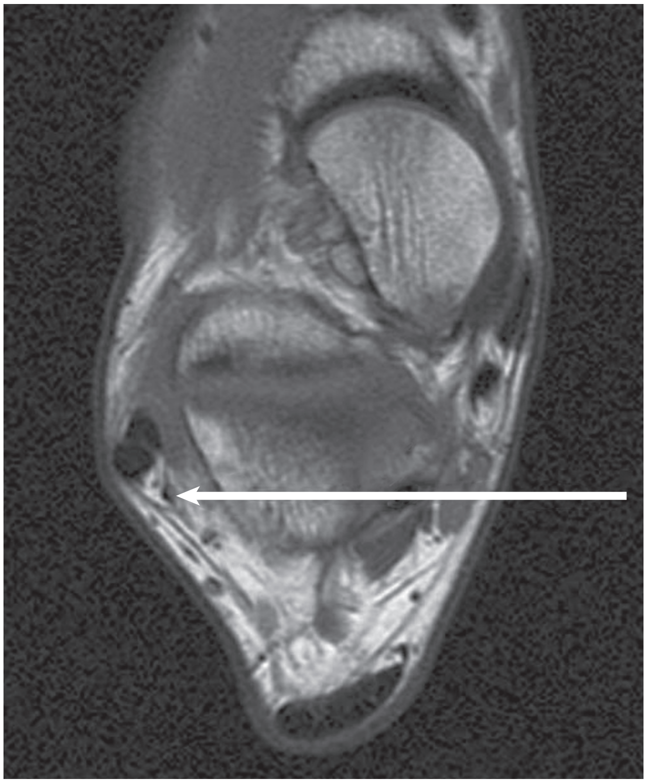

Fig. 112.13 A T1-weighted axial image of the ankle highlighting the presence of a peroneus quartus *(arrow)*. The tendon is a small-caliber low signal intensity structure located posterior within the peroneal retinaculum. Proximally the tendon has its own muscle belly and typically will insert on the calcaneus. (From Joos D, Kadakia AR. Peroneal tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

Direct magnetic resonance imaging signs. As measured on axial imaging, abnormal thickening of the peroneal tendon is noted if the tendon has a larger diameter than the posterior tibial tendon. Intermediate signal intensity within the peroneal tendons noted on three consecutive T1-weighted proton density–weighted images is 92% sensitive for the detection of clinically relevant peroneal tendon disorders (Fig. 112.14).[7] Fluid within the peroneal tendon sheath noted on T2-weighted imaging is a nonspecific finding that can be seen commonly in asymptomatic persons. However, the presence of circumferential fluid within the peroneal tendon sheath of greater than 3 mm in width is highly suggestive of clinically relevant peroneal tenosynovitis. The presence of a bisected or split peroneus brevis on either T1- or T2-weighted imaging is diagnostic of a tear (Fig. 112.15). A split that does not extend across the full width of the tendon is a partial tear.

Indirect magnetic resonance imaging signs. Irregular contour of the tendons on either T1- or T2-weighted imaging is suggestive of a peroneal tendon tear. The presence of a "C-shaped" peroneus brevis with the arms extending posteriorly around the peroneus longus is consistent with a tear. Thickening or attenuation of the lateral collateral ligaments warrants careful examination of the peroneal tendons because the two conditions commonly coexist. A peroneal tubercle that is greater than 5 mm

Fig. 112.14 A T1-weighted axial image with intermediate signal intensity in the location of the peroneus brevis. This finding was noted on multiple axial images, consistent with severe tendinosis of the peroneus brevis *(arrow)*. Obvious thickening of the tendon is also noted. The normal peroneus longus is noted posterior to the pathologic brevis. (From Joos D, Kadakia AR. Peroneal tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

Fig. 112.15 A T2-weighted axial image of a longitudinal split tear of the brevis *(arrowheads)*. The normal peroneus longus *(arrow)* lies posterior to the split brevis. (From Joos D, Kadakia AR. Peroneal tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

is an uncommon finding in normal persons and can be associated with peroneus longus synovitis and rupture. A "low-lying" muscle belly of the peroneus brevis is associated with peroneal tendon disease. Although the muscle belly can extend to the tip of the fibula in one third of normal persons, it rarely extends 1 cm past the fibular tip.

Pitfalls

Increased fluid signal within the tendons can be noted as they course around the distal fibula and is termed the *magic angle phenomenon*.[8] This signal is normal and should not be confused with a pathologic finding. The presence of intermediate signal intensity or fluid is a very common finding in healthy persons. MRI alone is specific for the diagnosis of peroneal tendon disease and must be used only as an adjunct with a good clinical examination to prevent misdiagnosis. The presence of a peroneus quartus should not be mistaken as a split tear of the peroneal tendons. The tendon has its own muscle mass, and the tendon structure of the longus and the brevis will be normal. Bifurcated or trifurcated tendon slips can be differentiated from a tear by the presence of the slip within the muscle proximal to the ankle joint.

Anterior Tibial Tendon

MRI and dynamic ultrasound are the two most common imaging studies used to confirm the diagnosis of a rupture of the tendon. MRI is excellent to determine the extent of disease in a patient with insertional tendinosis. If the clinical situation is obvious based on the history and physical examination, further imaging is not necessary. However, if surgical intervention is planned, these studies are useful for preoperative planning.

Magnetic Resonance Imaging

Direct magnetic resonance imaging signs. Discontinuity of the tendon is consistent with a complete tear of the tendon (Fig. 112.16). Intermediate signal intensity within an extensor tendon noted on consecutive T1-weighted proton density–weighted images with an increased tendon diameter is sensitive for tendinosis (Fig. 112.17).[9] Tendon thickness of greater than 5 mm within the distal 3 cm of the anterior tibial tendon is 94% sensitive in the diagnosis of tendinosis. Increased signal intensity on both T1- and T2-weighted imaging without discontinuity of the tendon is diagnostic of a partial tear.[9]

Indirect magnetic resonance imaging signs. Dorsal osteophyte formation along the midfoot or hindfoot can be associated with an attritional anterior tibial tendon tear.

Pitfalls

Review of only T1-weighted imaging may lead to a misdiagnosis of a ganglion as tendinosis, because it will appear as intermediate signal intensity. This appearance is easily differentiated from tendinosis on T2-weighted imaging by the high signal intensity (fluid) of the ganglion that is not seen with tendinosis.

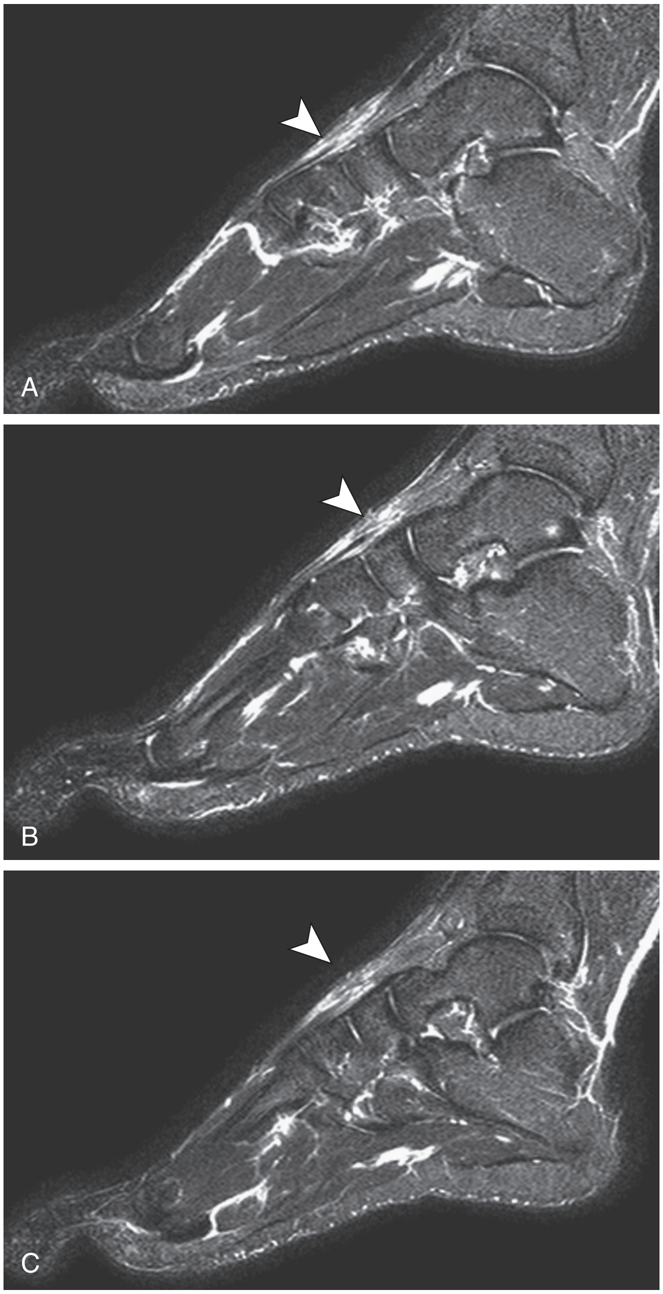

Fig. 112.16 Three consecutive sagittal T2-weighted images in a patient with a laceration of the extensor hallucis longus *(EHL)*. The EHL distal stump is identified (A, *arrowhead*). (B) The site of rupture *(arrowhead)*; note the rounding off of the cut edge. (C) Edema *(arrowhead)* and lack of the tendon that has retracted proximally. (From Joos D, Kadakia AR. Extensor tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

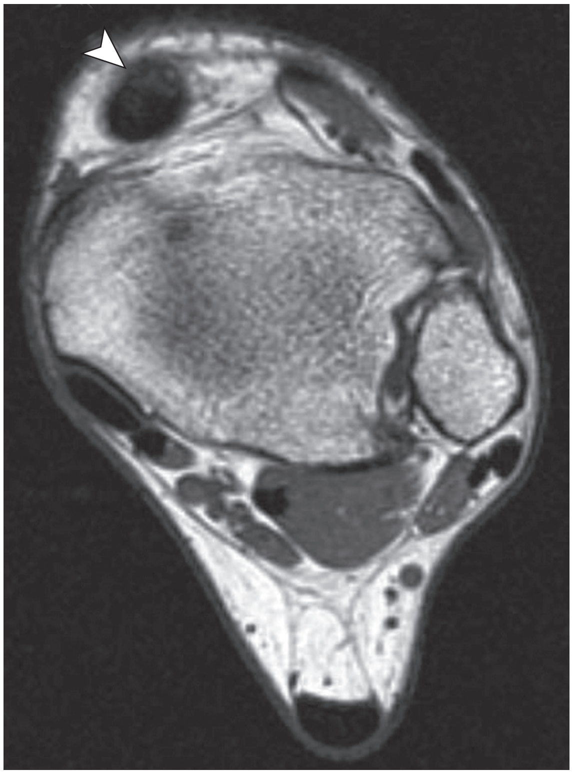

Fig. 112.17 An axial T1-weighted image of a patient with severe tendinosis of the anterior tibial tendon *(arrowhead)*. The tendon is thickened with significant intermediate signal present that is obliterating the normal architecture of the tendon. (From Joos D, Kadakia AR. Extensor tendon disorders. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

IMAGING OF LIGAMENTS

Magnetic Resonance Imaging

Normal Appearance

The ankle ligaments are uniformly low in signal intensity on all imaging sequences. The anterior talofibular ligament is best seen in the axial plane on proton density and proton density fat-saturated images as a thin linear structure (Fig. 112.18). The calcaneofibular ligament (CFL) is well visualized in the axial and coronal planes. The syndesmotic ligaments are best seen in the axial plane. The deltoid ligament is best seen in coronal plane.

Direct Magnetic Resonance Imaging Signs

Acute ligamentous injuries are classified by MRI as grade I, II, or III. Grade I injuries will demonstrate edema (but not discrete fluid) around the ligament. Grade II injuries will have increased signal within the ligament, consistent with a partial-thickness tear. Fluid can also be present around the ligament. In the setting of a grade III injury, complete disruption of the ligament will be noted by the absence or discontinuity of the ligament. Avulsion fractures may be seen at the site of ligament insertion.

Remote or chronic ligament injuries, sprains, and tears have characteristic appearances on MRI. The previously injured ligament may be increased in signal, wavy, thickened, and attenuated or absent (see Figs. 112.7 and 112.8).[10] Typically no soft tissue edema surrounds the remotely torn ligament, which helps one distinguish between an acute and chronic injury.

Not only is MRI accurate at evaluating the ankle ligaments, but it can also help rule out or occasionally rule in other pathologic conditions. Osteochondral lesions of the talus (OLTs) and syndesmotic injuries can be difficult to distinguish from post-sprain instability. MRI is quick to identify most of these confounding processes.

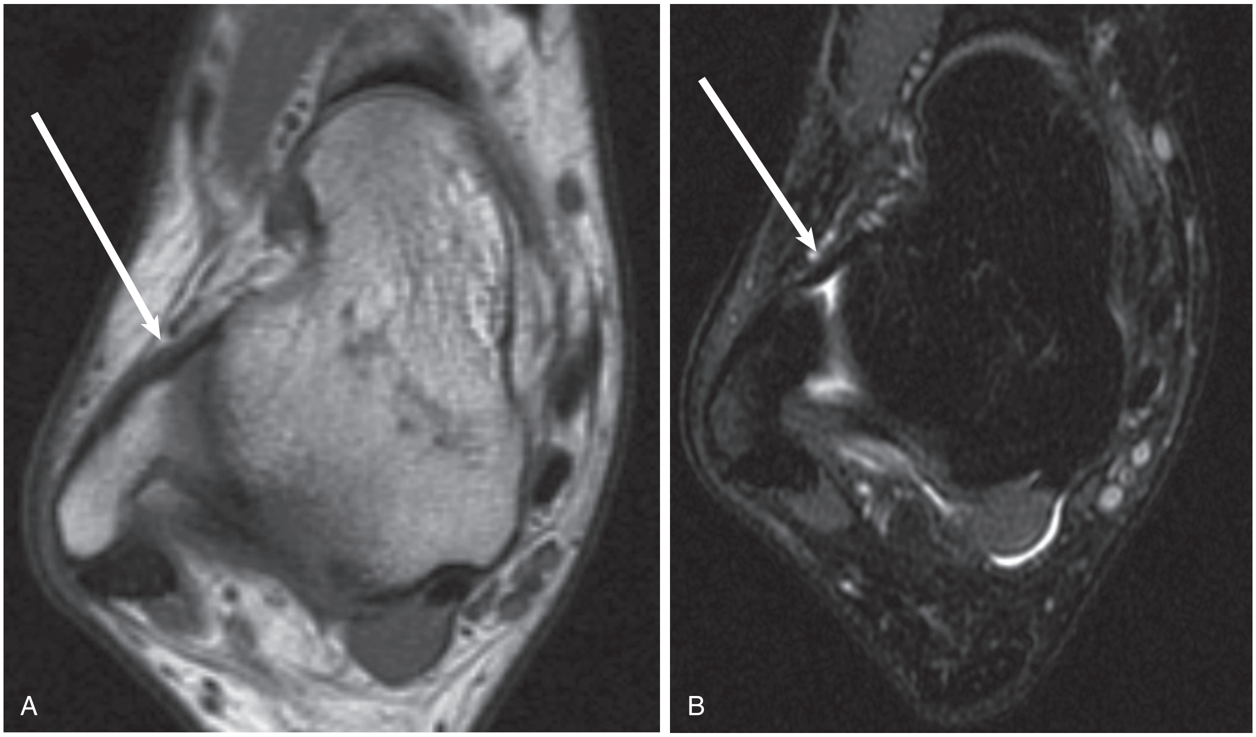

Fig. 112.18 Axial T1-weighted (A) and T2-weighted fat-saturated (B) images demonstrating the anterior talofibular ligament *(arrow)*. The ligament is low in signal intensity, of uniform thickness, and taut. (From Joos D, Sabb B, Tran NK, et al. Acute ankle ligament injuries. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

Acute syndesmotic injury will demonstrate high signal intensity on T2-weighted imaging around or in the ligament with possibility of ligament disruption. Presence of high signal on T2-weighted imaging at origin of posterior inferior tibiofibular ligament indicates high-grade injury. Chronic injury may demonstrate wavy or curved ligamentous disruption with the possibility of irregular thickening without high signal intensity on T2-weighted imaging.

Indirect Magnetic Resonance Imaging Signs

Indirect syndesmotic injury may be noted by a linear fluid signal 1.25 cm/1.5 cm on T2-weighted fat-saturated images consistent with injury. This finding is not pathognomonic but is highly suggestive in the setting of trauma. The presence of an ankle effusion, especially when joint fluid leaks out of the joint capsule, is indicative of a tear of the anterior talofibular ligament. Bone marrow edema of the distal fibula or talar insertion of the anterior talofibular ligament can be present. Bone marrow edema of the medial malleolus and talar insertion of the deltoid ligament may also be seen.

Pitfalls

Fluid in the peroneal tendon sheath can be a secondary sign of CFL injury; however, peroneal tenosynovitis can incite surrounding edema (peritendinitis) and result in the appearance of a CFL sprain (a pseudosprain of the CFL). A couple of important exceptions exist to the rule that the ligaments are dark on all conventional MRI sequences. In particular, the deltoid ligament and the posterior talofibular ligament are usually intermediate to bright on T2-weighted fat-saturated images. This normal appearance can be misinterpreted as an acute or chronic ligament sprain. Meniscoid lesions can be easily overlooked on imaging studies. One must be vigilant and systematic in the evaluation of the imaging to correctly establish this diagnosis by imaging. The lesion can be a source of pain for the patient and should be debrided if surgical intervention is performed.

OSTEOCHONDRAL LESIONS

OLTs may be visualized on plain radiographs as a result of an associated fracture of the subchondral bone or a cystic defect. However, these findings are not universally present and thus in cases of chronic injuries with mechanical symptoms, such as locking or giving way, or a feeling of instability of the ankle joint, in addition to pain and persistent swelling, MRI is the imaging modality of choice to visualize the articular surface and subchondral bone. However, caution is necessary because the presence of abnormal findings on MRI, especially within the medial talar dome, may not be the causative factor for the patient's complaint. Combining the information from the history and physical examination along with the MRI findings is critical prior to recommending surgical intervention for an OLT.

Magnetic Resonance Imaging

Normal Appearance

The talus is normally bright on T1-weighted images and is dark on the T2-weighted fat-saturated images. Because the cartilage of the talus is thin, it is sometimes difficult to see on conventional MRI sequences. However, when an osteochondral lesion is present, the effect on the subchondral bone is very apparent. MRI allows multiplanar imaging of the talus for localization and

Fig. 112.19 Sagittal magnetic resonance imaging demonstrating the most common central location of osteochondral lesions. The lesion has the typical dark appearance on T1-weighted imaging. (From Joos D, Sabb B, Kadakia AR. Osteochondral lesions. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

characterization of OLTs. MRI has been proven itself sensitive and specific in the diagnosis of OLTs.

Direct Magnetic Resonance Imaging Signs

The lesions can be located anywhere in the talar dome (but most often in the lateral or medial aspect of the talus, in its midportion from anterior to posterior; Fig. 112.19). The lesions are typically dark on T1-weighted images and variable on T2-weighted fat-saturated images. Signs of instability include fluid undercutting the lesion, a cystic lesion, or a high-intensity T2-weighted fat-saturated signal at the interface between the lesion and the underlying talus and partial or complete separation of the fragment from its normal location (Fig. 112.20).[11] Cystic lesions appear as a bright fluid-filled area beneath the subchondral bone on T2-weighted imaging.

Indirect Magnetic Resonance Imaging Signs

Intra-articular bodies must be scrutinized, and one would have to assess for talar dome donor sites. One of earliest signs of OLT is decreased T1-weighted signal, often without significant increased T2-weighted fat-saturated signal or any cartilage defect.[11]

Pitfalls

Early OLT may only appear as low T1-weighted signal intensity and should not be confused with posttraumatic edema or early osteoarthritis (OA). OA can also result in osteochondral abnormalities in the talus. This process is separate and occurs by a different mechanism. One pitfall is to have a patient with OA mistakenly diagnosed as having OLT; the treatment of osteochondral lesions in OA is not the same as the treatment of OLT in a younger nonarthritic patient. One way to help make the distinction is to evaluate the joint for overall changes of OA, including osteophyte, joint space narrowing, and changes of the associated tibial plafond.

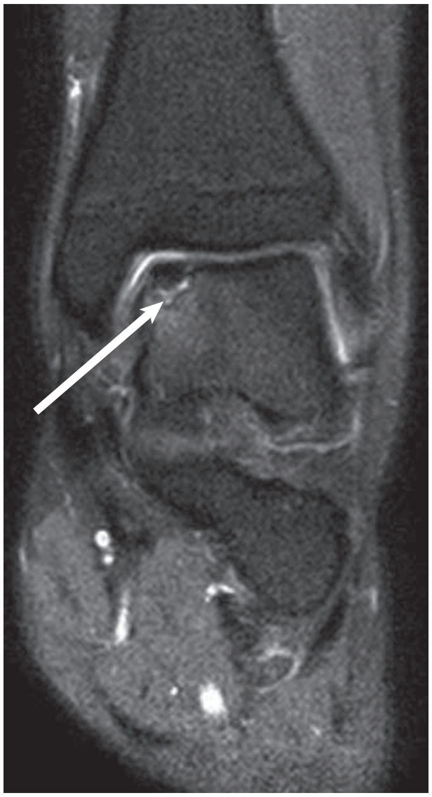

Fig. 112.20 A patient with an unstable osteochondral lesion of the talus. The bright fluid undercutting the lesion is well visualized on the fluid-sensitive T2-weighted fat-saturated image *(arrow)*. Fluid interposing between the osteochondral lesion of the talus and the talus is an excellent indicator of instability. (From Joos D, Sabb B, Kadakia AR. Osteochondral lesions. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

NERVE ENTRAPMENT

Entrapment or compression of the peripheral nerves at the foot and ankle are common and often unrecognized causes of pain and disability. These disorders are diagnosed primarily by clinical examination and adjunctive nerve testing if such testing is believed appropriate. The use of ultrasound has been advocated by some authors in the diagnosis and as a localizing aid for injection of Morton neuromas. The primary role of MRI in the treatment of nerve compression disorders is to rule out a mass-occupying lesion, most commonly in the setting of suspected tarsal tunnel. Routine use of MRI in the evaluation of nerve disorders is not advocated.

Magnetic Resonance Imaging: Interdigital Nerve

Normal Appearance

The nerve should appear circular and less than 3 mm in width.

Direct Magnetic Resonance Imaging Signs

Coronal T1-weighted imaging is most useful for evaluation of the presence of a Morton neuroma. Use of contrast-enhanced T2-weighted imaging can be used to ensure that the mass is a neuroma, differentiating it from other lesions. The neuroma should appear as an ovoid or dumbbell-shaped plantar mass (inferior to the intermetatarsal ligament) between the metatarsal heads (Fig. 112.21). Neuromas will appear with low to

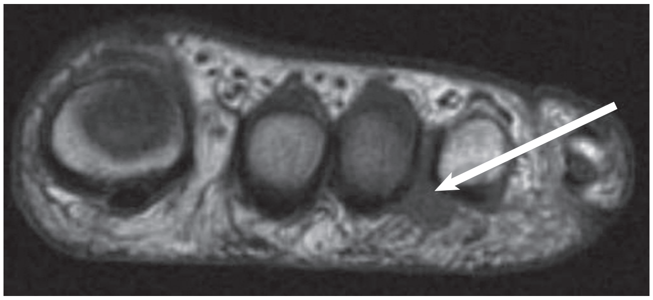

Fig. 112.21 A coronal T1-weighted image of a Morton neuroma *(arrow)*. Note that the neuroma is pear shaped and is plantar to the intermetatarsal ligament. (From Seybold J, Kadakia AR. Nerve entrapment syndromes. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

intermediate signal intensity on T1- and T2-weighted imaging. Although MRI can aid in the diagnosis of a neuroma, the sensitivity has been reported as 76% to 87% with surgical confirmation as the gold standard. Surgically confirmed neuromas have been shown to have a high association with widths greater than 5 mm.

Indirect Magnetic Resonance Imaging Signs

The neuroma must be visualized directly to be diagnosed.

Pitfalls

Intermetatarsal bursitis is differentiated from a neuroma by its small size (<3 mm) and high signal intensity on T2-weighted imaging, consistent with fluid.

Magnetic Resonance Imaging: Tarsal Tunnel

Normal Appearance

The normal appearance of the tarsal tunnel should demonstrate the contents of the tarsal tunnel without any evidence of a mass-occupying lesion. The normal contents of the tarsal tunnel will be noted with a thin overlying flexor retinaculum. Axial imaging provides the most information (Fig. 112.22).

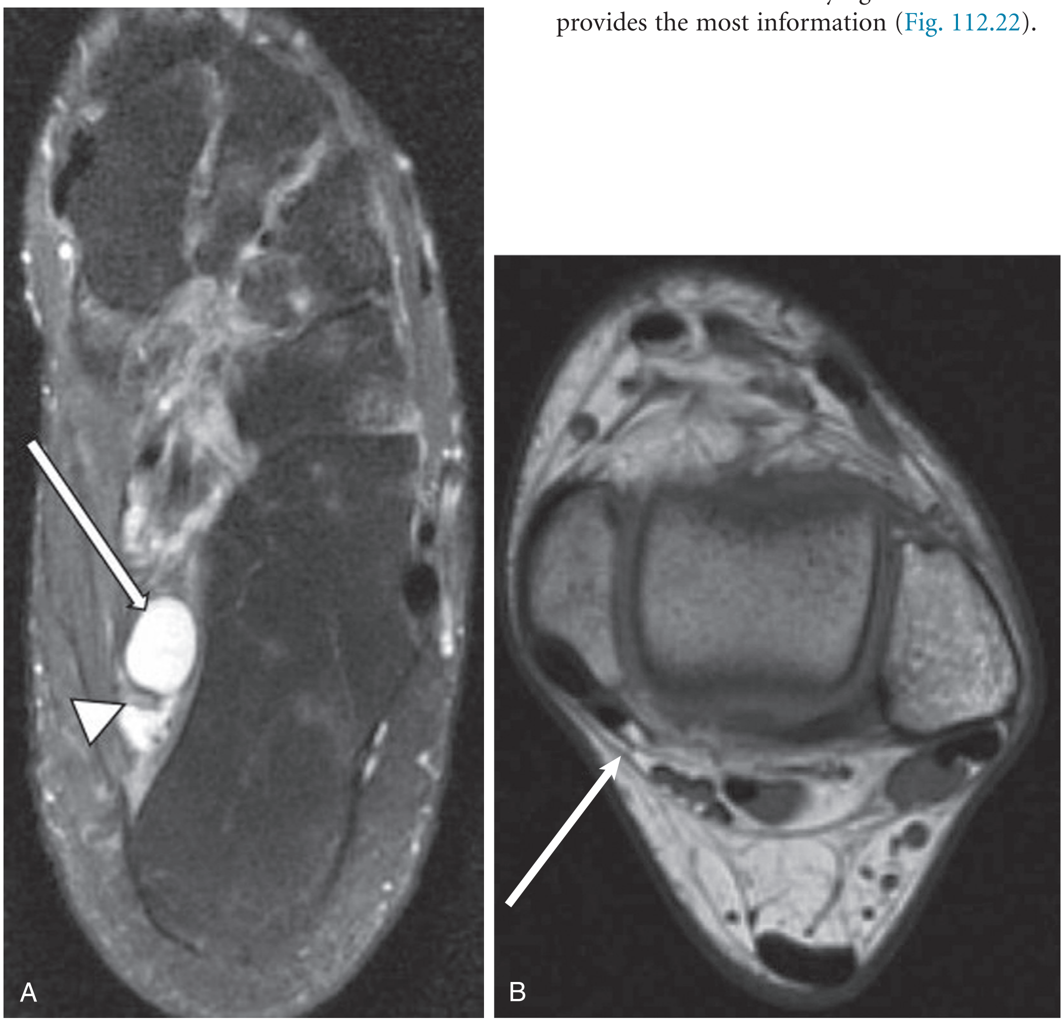

Fig. 112.22 (A) T2-weighted fat-saturated image of ganglion cyst *(arrow)* in tarsal tunnel compressing tibial nerve *(arrowhead)*. (B) An axial T1-weighted image of a normal tarsal tunnel. The flexor retinaculum can be visualized *(arrow)* as a thin hypoechoic band. The contents of the tarsal tunnel are easily visualized without any mass-occupying lesion. (From Seybold J, Kadakia AR. Nerve entrapment syndromes. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

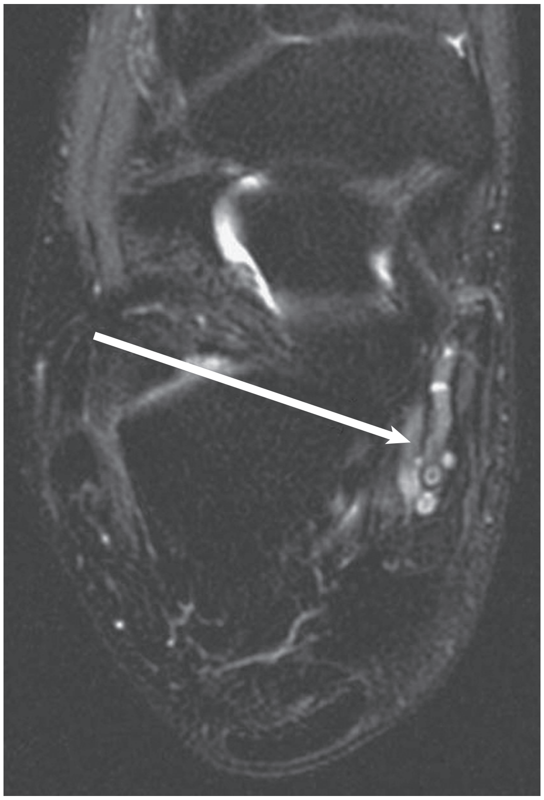

Fig. 112.23 An axial T2-weighted fat-saturated image of venous varicosities *(arrow)* within the tarsal tunnel. These varicosities appear as torturous high signal intensity structures. (From Seybold J, Kadakia AR. Nerve entrapment syndromes. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

Direct Magnetic Resonance Imaging Signs

The most common MRI finding in the setting of tarsal tunnel is a mass-occupying lesion. These lesions include neurofibrosarcoma, neurilemoma, ganglion (see Fig. 112.22A), hemangioma, venous varicosities or dilated posterior tibial veins (Fig. 112.23), FHL tenosynovitis with fluid within the tendon sheath, and hypertrophy of the abductor hallucis.

Indirect Magnetic Resonance Imaging Signs

Flattening of the tibial nerve suggests tarsal tunnel syndrome, and a mass-occupying lesion should be sought.

PLANTAR FASCIITIS

MRI is not required to obtain the diagnosis of either planar fasciitis or fibromatosis. The primary use is to rule out other conditions such as a calcaneal stress fracture, plantar fascia rupture, neoplasm, and Baxter neuritis.

Magnetic Resonance Imaging

Normal Appearance

Sagittal and coronal imaging is superior for visualization of the plantar fascia. The ligament should appear hypoechoic on all sequences, and the maximum thickness is no greater than 4 mm.

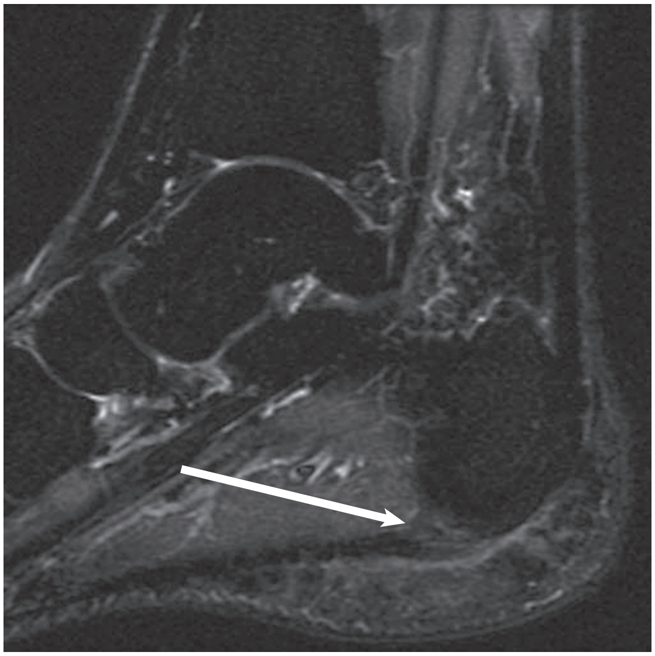

Fig. 112.24 A sagittal T2-weighted image demonstrating increased signal at the origin of the plantar fascia *(arrow)* in a patient with plantar fasciitis. (From Seybold J, Kadakia AR. The plantar fascia. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions.* Philadelphia: Elsevier; 2011.)

Direct Magnetic Resonance Imaging Signs

Thickening of the plantar fascia (7 to 8 mm) at the insertion is seen with fasciitis. Increased signal intensity at the insertion of the fascia is seen best on T2-weighted or short tau inversion recovery sequences (Fig. 112.24). Plantar fascia rupture is noted by high signal intensity (fluid) at the proximal aspect of the plantar fascia with complete discontinuity from the calcaneal origin. Plantar fibromatosis can be identified by a single or multiple subcutaneous nodules that are not commonly greater than 3 cm in diameter.

Indirect Magnetic Resonance Imaging Signs

Increased signal intensity at the calcaneal insertional site is suggestive of plantar fasciitis, along with surrounding subcutaneous edema. With plantar fibromatosis, no reactive edema is usually noted.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Recht MP, Donley BG. Magnetic resonance imaging of the foot and ankle. *J Am Acad Orthop Surg*. 2001;9:187–199.

Level of Evidence:
V

Summary:
The authors provide an excellent overview of the normal and pathologic findings noted with magnetic resonance imaging of

the foot and ankle. The inclusion of excellent figures aids in the usefulness of this review.

Citation:
Feighan J, Towers J, Conti S. The use of magnetic resonance imaging in posterior tibial tendon dysfunction. *Clin Orthop*. 1999;365:23–38.

Level of Evidence:
III

Summary:
The authors present an excellent review of the appropriate technique and abnormalities noted on magnetic resonance imaging of posterior tibial tendon dysfunction. Supplemental cases provide additional clarification.

Citation:
Lo LD, Schweitzer ME, Fan JK, et al. MRI imaging findings of entrapment of the flexor hallucis longus tendon. *AJR Am J Roentgenol*. 2001;176:1145–1148.

Level of Evidence:
III

Summary:
The authors provide a retrospective review of the variable imaging findings noted in patients with flexor hallucis longus tendon entrapment. The authors provide a thorough discussion of the multiple causes for this disorder and their relevant imaging findings.

Citation:
Kijowski R, De Smet A, Mukharjee R. Magnetic resonance imaging findings in patients with peroneal tendinopathy and peroneal tenosynovitis. *Skeletal Radiol*. 2007;36:105–114.

Level of Evidence:
III

Summary:
The authors of this excellent study compare the magnetic resonance imaging findings of patients who have peroneal disease with the findings of persons who do not have any pathologic abnormality. The authors were able to conclude that intermediate signal was required on three consecutive images to be consistent with disease and that circumferential fluid of greater than 3 mm in maximal width is sensitive for peroneal tenosynovitis.

Citation:
Mengiardi B, Pfirrmann CW, Vienee P, et al. Anterior tibial tendon abnormalities: MR imaging findings. *Radiology*. 2005;235:977–984.

Level of Evidence:
III

Summary:
The authors of this excellent study compare the magnetic resonance imaging findings in patients who have known disease of the anterior tibial tendon with findings in age-matched control subjects. The authors concluded that thickening of greater than or equal to 5 mm and diffuse signal intensity abnormality within 3 cm of the insertion is consistent with anterior tibial tendinosis.

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